

ADITYA BAYU PRATAMA

Supervisor MES & IT Engineer | Smart Manufacturing & Industry 4.0

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PROFESSIONAL SUMMARY

Supervisor MES & IT Engineer with 7+ years designing and running Manufacturing Execution Systems, Industrial IoT and machine vision for high-precision electronics and oil & gas. Scaled the software and IT department from a solo role to five engineers, architected a custom C#/.NET MES governing 500+ product models, and cut a 5–7 hour manual feeder workflow to 30 minutes. Two-time 1st place winner of the Giken Global Value Engineering Award (2023, 2025).

CORE COMPETENCIES

Software & Languages: C# (.NET Framework), SQL/MySQL, Python, React.js, JavaScript, C++, LabVIEW, Dart/Flutter, REST APIs
Smart Manufacturing & Automation: MES architecture, Industrial IoT, traceability, Pokayoke, AOI machine vision, SMT process, PLC (BNSP certified), SCADA, Raspberry Pi, Arduino, data acquisition, PID control
AI & Modern Tooling: AI-assisted development, prompt engineering, LLM integration, LM Studio, Claude Code, Cursor, Codex, OpenCV/computer vision
Quality, IT & Leadership: FMEA, DMAIC, Six Sigma practices, 5M change management, value engineering, NPI, team management, department budget, network infrastructure, server maintenance, incident response

PROFESSIONAL EXPERIENCE

GIKEN PRECISION INDONESIA — BATAM, INDONESIA Apr 2019 – Present

Supervisor MES & IT Engineer Jul 2026 – Present

- Lead the MES and IT engineering function: system roadmap, architecture standards, project prioritisation, department budget and the engineering team.

Lead Software & IT Engineer Apr 2019 – Jun 2026

- Designed and deployed a custom Windows-based MES (C#/.NET, MySQL) governing 500+ product models across automotive, medical and consumer electronics lines, enforcing Pokayoke defect prevention and CTQ compliance.
- Built Giken One, an enterprise web platform (React.js, MySQL) digitising employee records, org charts, overtime approvals, IT ticketing and project scheduling.
- Engineered an automated ESD wristband gate (Arduino, C#, MySQL) enforcing 100% compliance before line entry, automated QR-code thermal label printing, and centralised file storage across all departments.
- Recruited, mentored and scaled the software and IT department from a solo role into a five-engineer team.
- Contained a worm malware outbreak across 20 laptops in 3 hours with no production downtime; sustained 99.9% uptime on core MySQL servers; resolved network outages inside 15-minute SLAs.

Software Test Engineer / New Product Introduction Engineer Apr 2019 – Apr 2024

- Built machine testing software and ran production trials for new electronic products; created FMEA, 5M change management, BOM and QCPC documentation for NPI.

CLADTEK BI-METAL MANUFACTURING — BATAM, INDONESIA Apr 2018 – Jan 2019

Junior Programmer — R&D Department Apr 2018 – Jan 2019

- Built an autonomous pipe-inspection robot (C#, Raspberry Pi, Lidar, light sensors) navigating inside industrial pipes, streaming live video and calculating weld defect location and severity.
- Developed low-latency data transmission protocols connecting embedded sensors to central servers for real-time welding QA evaluation.

SELECTED PROJECTS

Manufacturing Traceability System — Giken One MES (2019–present) — Plant-wide MES with per-line dashboards, full unit traceback and SMT refill control across 500+ product models. 1st place globally, Giken Value Engineering Award 2023.

Intelligent Machine Data (2025) — Real-time machine data acquisition and analytics replacing manual reporting with live line-efficiency and downtime visibility. 1st place globally, Giken VE Award 2025.

Auto Feeder Machine Data Generator (2026) — Digitalised the SMT feeder data flow across four roles: 5–7 hours with 4–6 people became a 30-minute automated flow, cutting cycle time by roughly 93%.

PCBA Management Apps (2026) — React.js and MySQL platform running the PCBA IT department: tickets, tasks, schedules, projects, attendance, assets and knowledge base on one dashboard.

EDUCATION

STATE POLYTECHNIC OF BATAM Jul 2017 – Jul 2021
Bachelor of Applied Science (S.Tr.T.), Mechatronics, Robotics & Automation Engineering — Graduated Cum Laude, GPA 3.81/4.00

HONORS & CERTIFICATIONS

Awards: 1st Place, Giken Global Value Engineering Award — 2025 (Intelligent Machine Data) and 2023 (Manufacturing Traceability System).

Certifications: Google Data Analytics Professional Certificate (Coursera, 2022) · PLC Competency, BNSP Indonesia (2019) · Electrical Power Installation Technician, LPJK (2020) · DevOps, Computer Networks, Dart and Flutter (Dicoding, 2023)